HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and implemented novel **liquid and lyophilised formulations** for a live virus vector vaccine, focusing on enhancing thermal stability and shelf-life for broad deployment.
- Executed a research program focused on drying processes and **thermostabilisation** for vaccine candidates, ensuring suitability for scalable manufacturing and subsequent preclinical testing.
- Collaborated with cross-functional teams, including manufacturing partners and the VITAL group, to identify and verify formulation strategies, contributing to the advancement of a Lassa fever vaccine.
- Prepared and presented detailed research reports on formulation development and **process validation** findings to project partners and the Horizon-EU funding body.
- Conducted **pharmaceutical and virology research** methodologies, applying practical problem-solving skills to overcome challenges in vaccine stabilization and deployment.
- Supported and mentored early-career researchers within the VITAL group, fostering a collaborative environment for advancing vaccine technology and access initiatives.
- Managed and analyzed **production data** from enterprise systems using **SQL** and **Python scripting**, contributing to operational dashboards and reporting for financial services clients.

SKILLS

Pharmaceutical Sciences – Vaccine Formulation, Thermal Stability, Lyophilisation, Liquid Formulations, Preclinical Testing, Scalable Manufacturing, Process Validation

Research & Development – Research Program Design, Data Analysis, Report Preparation, Presentation, Problem Solving, Record Keeping, Mentoring

Programming & Scripting – Python, SQL, Java, Spring Boot, REST APIs

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, pandas, NumPy, scikit-learn, R / RStudio

Enterprise Systems – ERP Integration Support, Production Support, SLA Management, Incident Response, Change Management

PROJECTS

Development of Thermostable Vaccine Formulations

- Engineered novel **liquid and lyophilised formulations** for a live virus vector vaccine candidate, aiming to extend its operational stability beyond ultra-cold chain requirements.
- Applied **pharmaceutical formulation techniques** and conducted stability studies to evaluate the impact of excipients and processing parameters on vaccine efficacy.
- Documented formulation development processes and results, preparing technical summaries for internal review and potential transfer to manufacturing partners.

Automated Data Pipeline for Operational Reporting

- Developed a **Python scripting**-based data pipeline to extract and transform data from multiple enterprise databases (**PostgreSQL**, **MySQL**) for a financial services operations dashboard.
- Automated the generation of weekly operational reports using **SQL queries** and data aggregation in **pandas**, reducing manual reporting effort by approximately 15 hours per week.
- Implemented data validation checks within the pipeline to ensure accuracy and integrity of metrics displayed on the dashboard, supporting transaction monitoring workflows.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)